EXPERIMENT 3

Title: Morphological Analysis Using Python

Aim:

To perform basic morphological analysis on English words using Python and the Natural Language Toolkit (NLTK). Morphological analysis is the process of studying the internal structure of words, including roots, prefixes, suffixes, and word forms. Stemming is a basic NLP technique that reduces words to their stem or base-like form.

Algorithm:

1. Start the program.
2. Import the PorterStemmer class from the nltk.stem module.
3. Create an object of the Porter Stemmer.
4. Define a list of words containing different forms, such as verbs, nouns, and words with suffixes.
5. Traverse the list of words one by one.
6. Apply the stem() method to each word.
7. Obtain the stem produced by the stemming algorithm.
8. Display each original word alongside its corresponding stem.
9. Compare the original words and their stems to observe how suffixes are reduced.
10. Stop the program.

Program:

Exp-03-Morphological Analysis Using NLTK.py - C:\Users\jerin\OneDrive\Desktop\NLP\Expe...

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```
from nltk.stem import PorterStemmer

stemmer = PorterStemmer()

words = ["playing", "played", "running",
         "studies", "cats", "connected"]

print("Morphological Analysis")

for word in words:
    stem = stemmer.stem(word)
    print(word, "->", stem)
```

Output:

```
>>>
= RESTART: C:\Users\jerin\OneDrive\Desktop\NLP\Experiments
Analysis Using NLTK.py
Morphological Analysis
playing -> play
played -> play
running -> run
studies -> studi
cats -> cat
connected -> connect
```

Result:

Thus, morphological analysis was performed successfully using Python and the NLTK library. The Porter Stemmer reduced several words to their stem forms by removing or modifying suffixes. This technique is useful in information retrieval, text classification, search engines, and other NLP applications. Stemming does not always produce a complete dictionary word,

as shown by studies -> studi.